

Scene understanding: Detecting objects and building relationship graphs

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Project goals

- Scene



Project goals

- Scene → Abstract representation



Project goals

- Scene → Abstract representation → Ask questions about contents



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- Scene → Abstract representation → Ask questions about contents



What is in this scene?

Project goals

- Scene → Abstract representation → Ask questions about contents



What is in this scene?

Where is the truck?

Project goals

- Scene → Abstract representation → Ask questions about contents



What is in this scene?

Where is the truck?

Is the snowman above the truck?

Related work

- [1] Generated scenes + fixed language grammar



The scene consists of three roads meeting at an intersection, a group of trees, an oak tree and three market stands. The three market stands are next to the first road. The group of trees consists of three pine trees and three bushes. The first market stand consists of a sign to the right of a table. A big pot of stew is in the middle of this table. The second market stand consists of a sign besides of a table. A big pot of stew is in the middle of this table. The third market stand consists of three flowerpots on top of a table and a sign. This sign is to the right of this table.

Related work

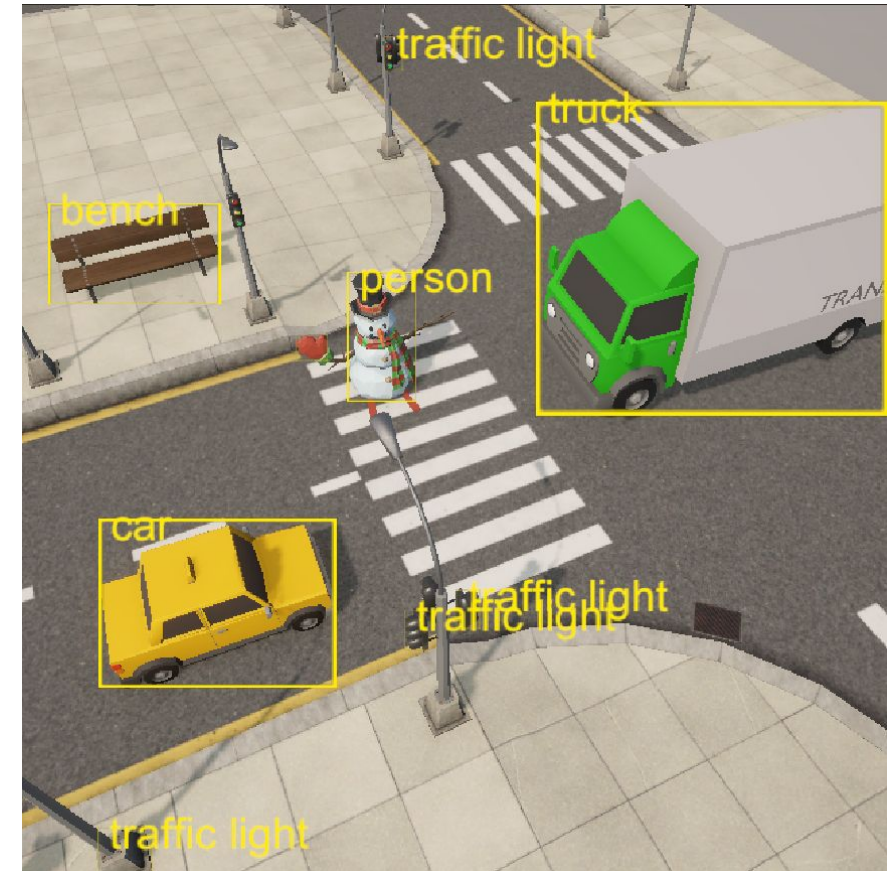
- [1] Generated scenes + fixed language grammar
- [2] RGB-D video + hand-crafted relationship graph

Related work

- [1] Generated scenes + fixed language grammar
- [2] RGB-D video + hand-crafted relationship graph
- [3] RGB-D image + relationship graph generated by neural network

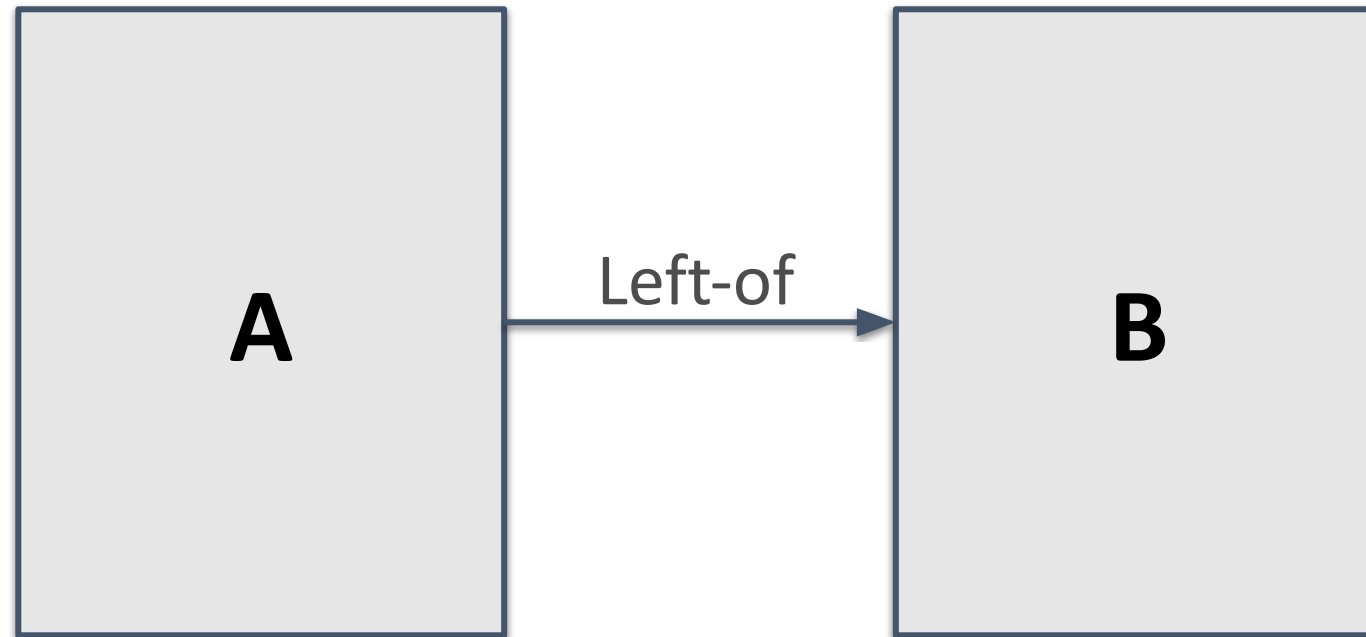
YOLO

- YOLO12s (s - small, model from HuggingFace)
- Integrated using Unity inference engine
- Camera frames rendered to texture and resized for model input
- Outputs class labels and bounding box coordinates



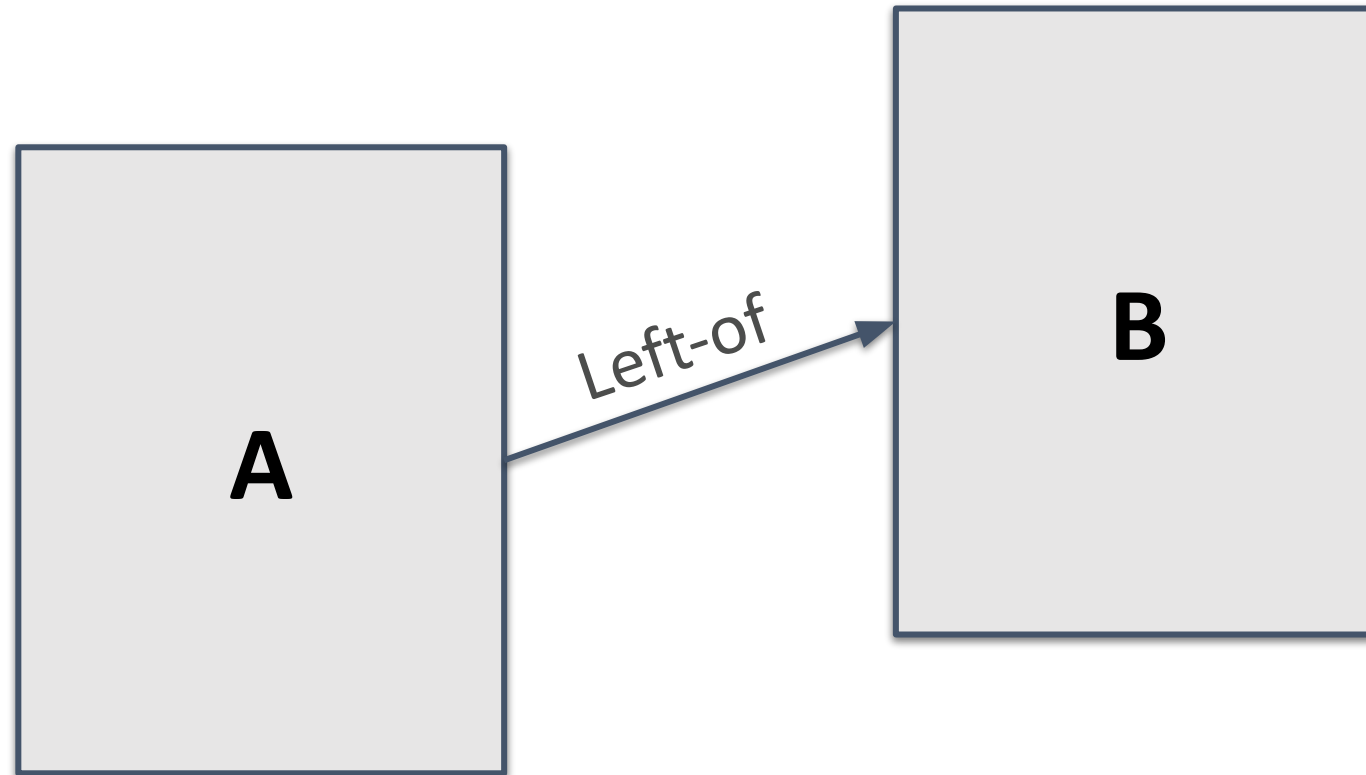
Relationship graph

- Idea: Take bounding boxes and analyze their positions and size



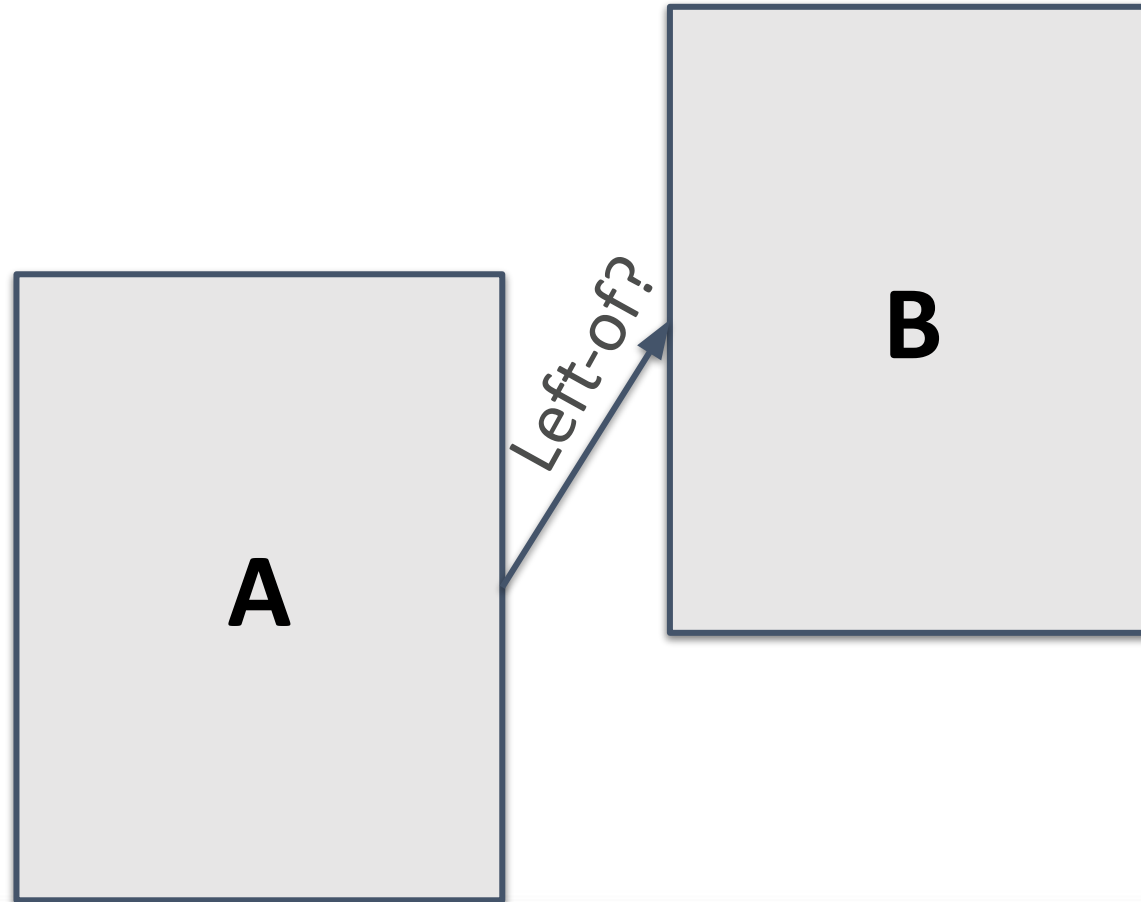
Relationship graph

- Idea: Take bounding boxes and analyze their positions and size



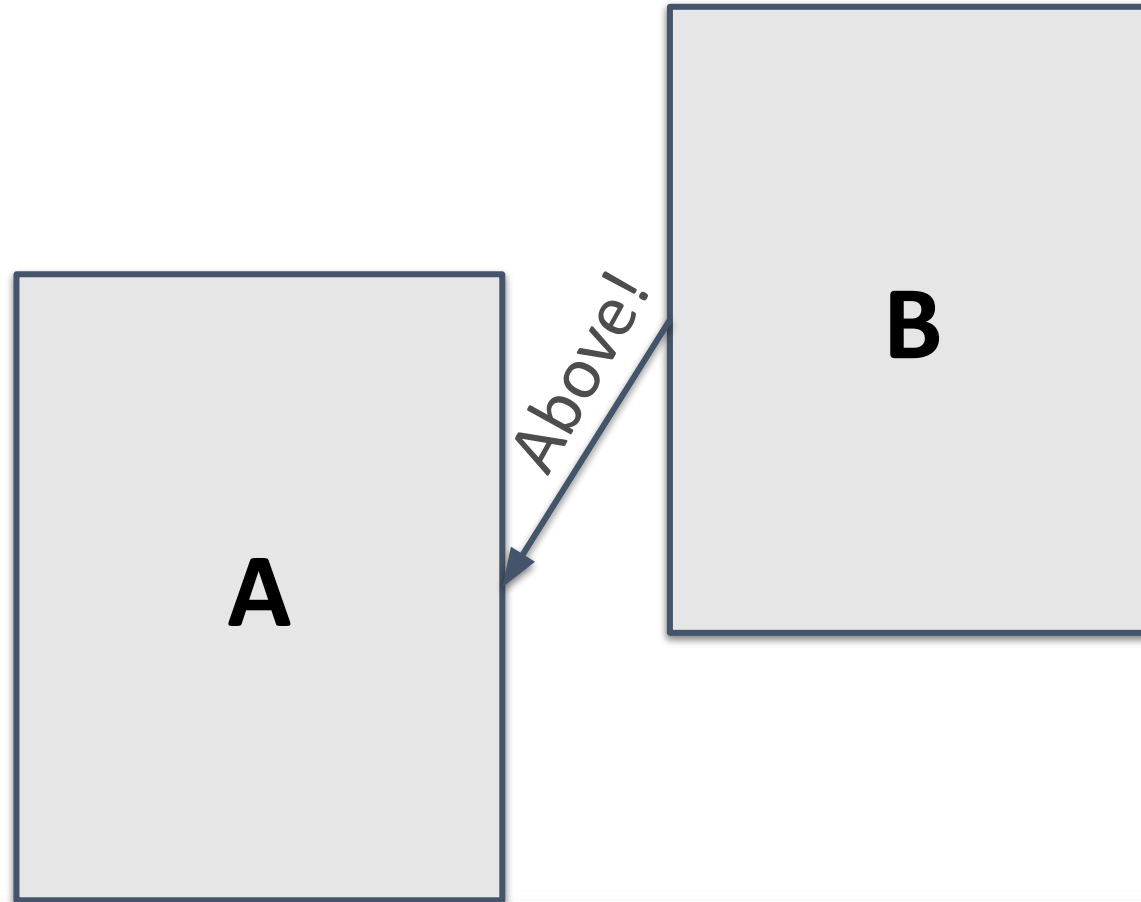
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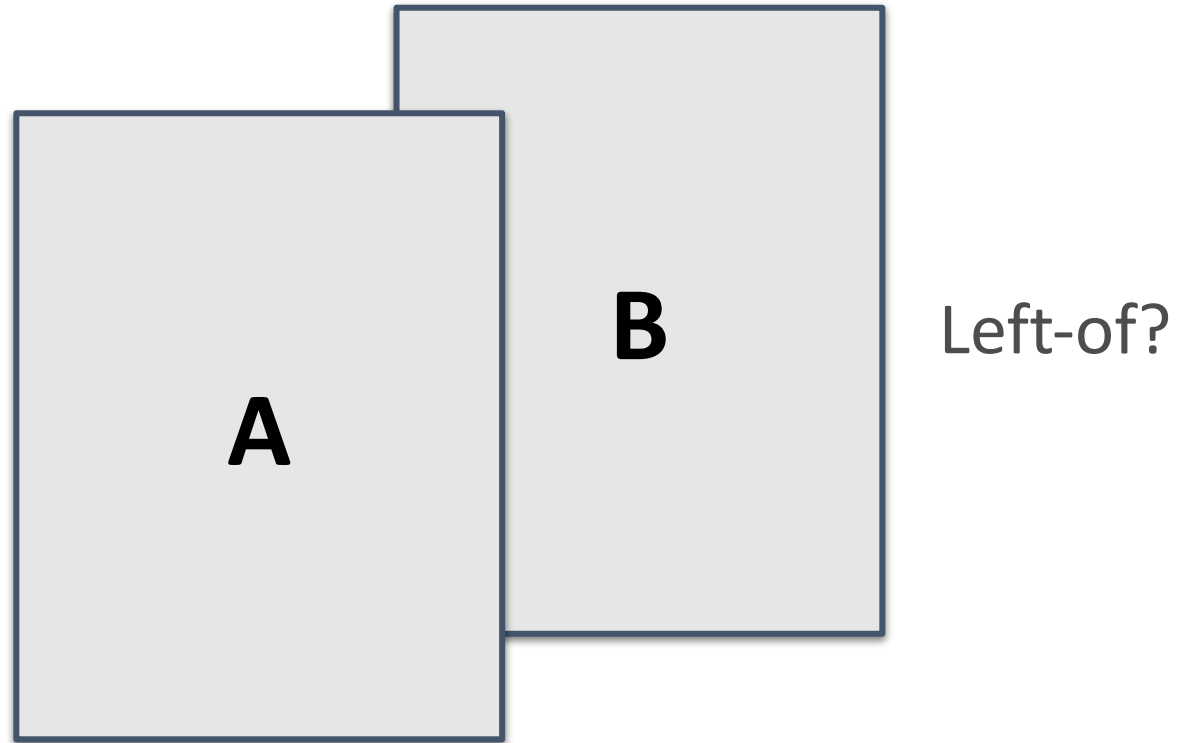
Relationship graph

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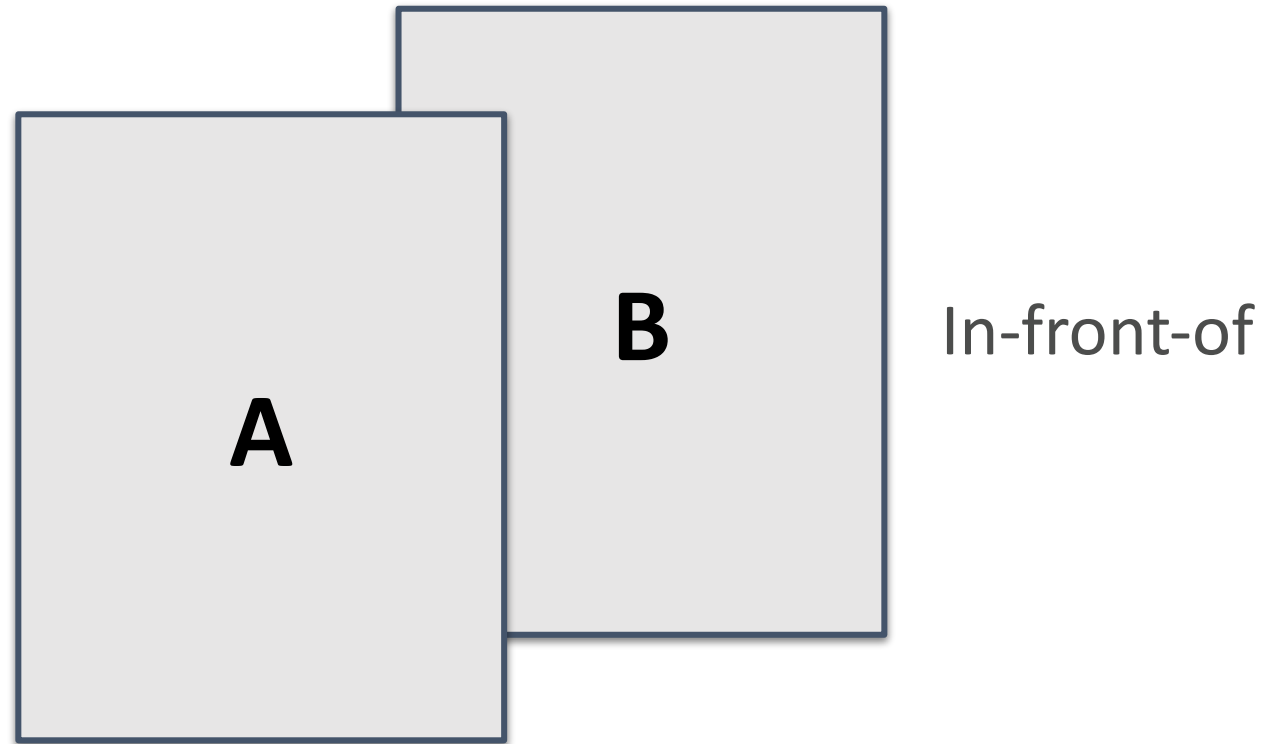
Relationship graph

- Idea: Take bounding boxes and analyze their positions and size



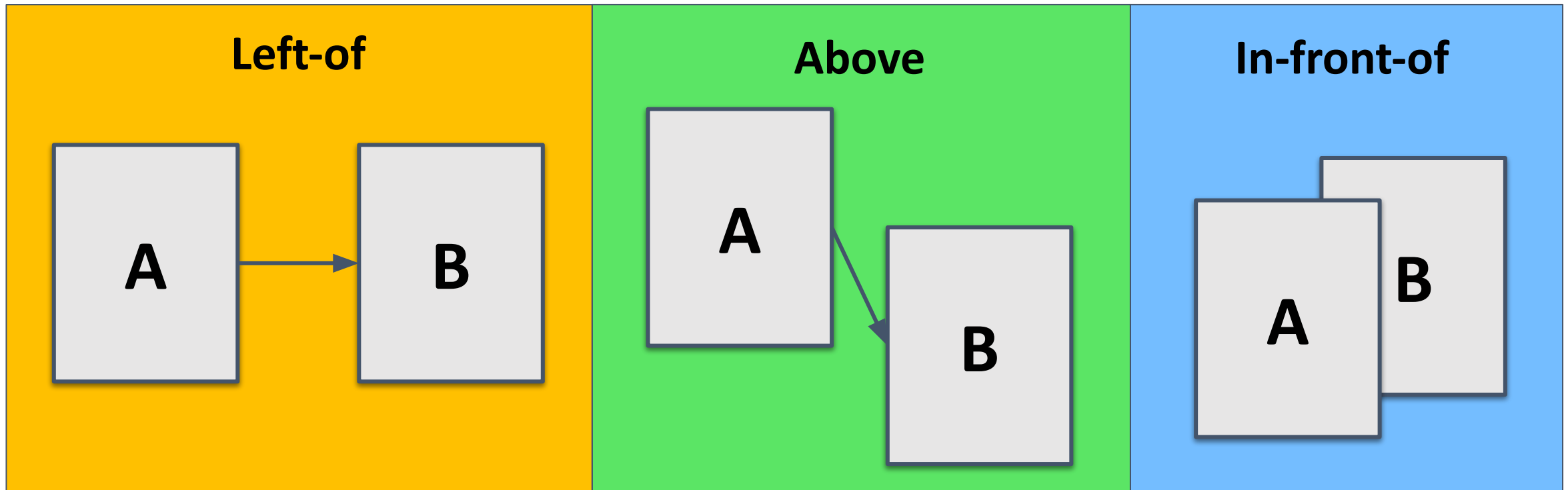
Relationship graph

- Idea: Take bounding boxes and analyze their positions and size



Relationship graph

- Idea: Take bounding boxes and analyze their positions and size
- Total of 3 relationships



Large language models

- Llama 3.2 1B (model HuggingFace)
- Integrated using UnityLLM
- Ask simple questions about the scene

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Prompt:

Consider the following graph. Pay attention to the order of the relationships: <graph in text form>. Fulfill the query in one sentence, human readable, short, and do not mention the graph: <user prompt>

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Graph:

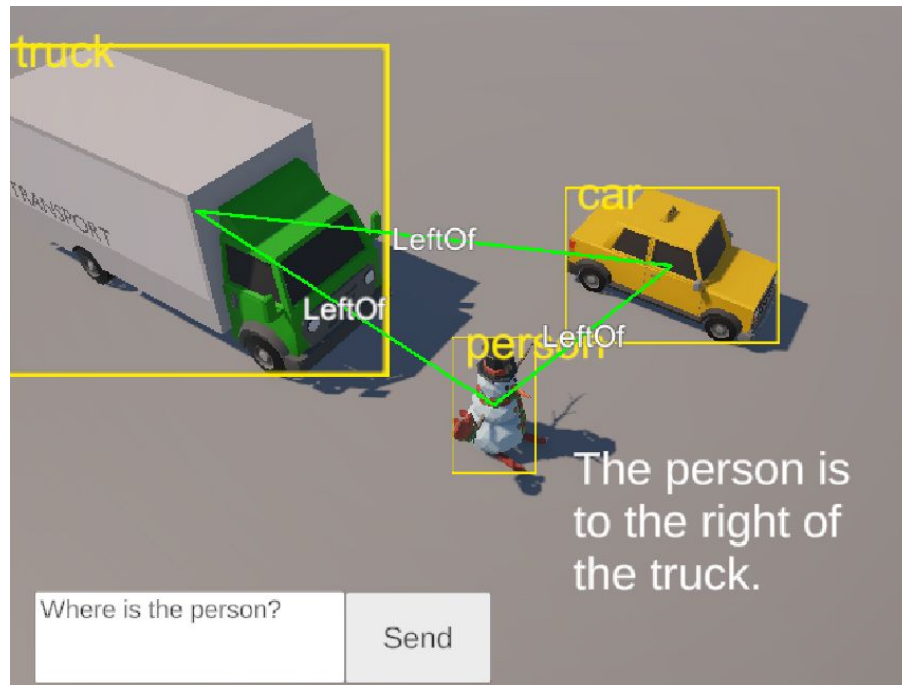
The person is to the left of the car

The truck is to the left of the car

The truck is to the left of the person

Large language models

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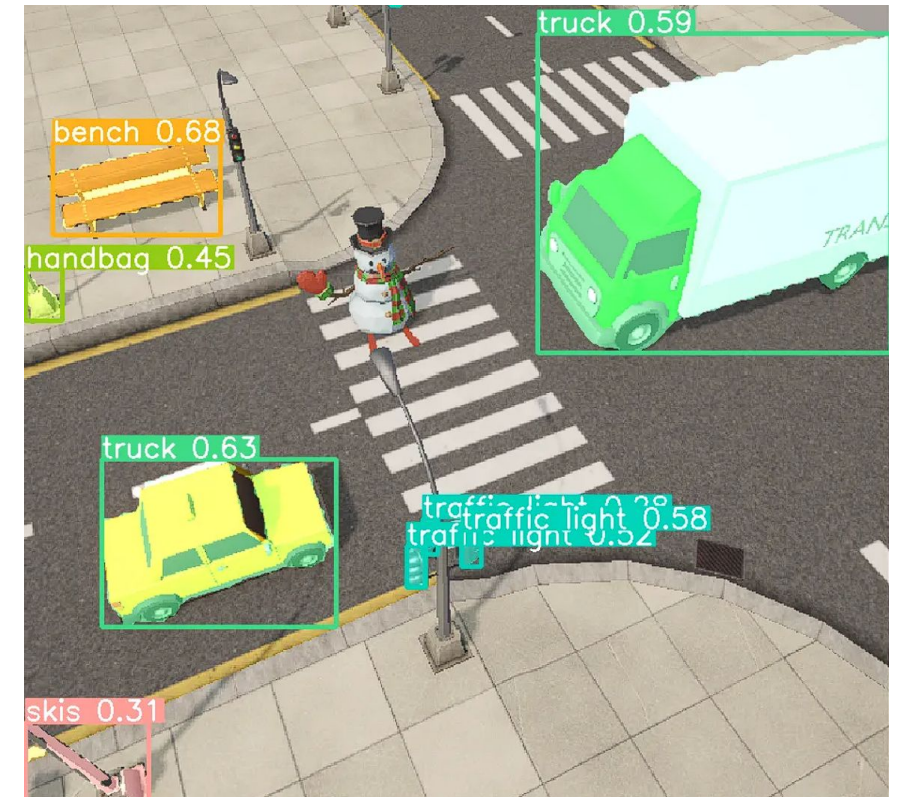
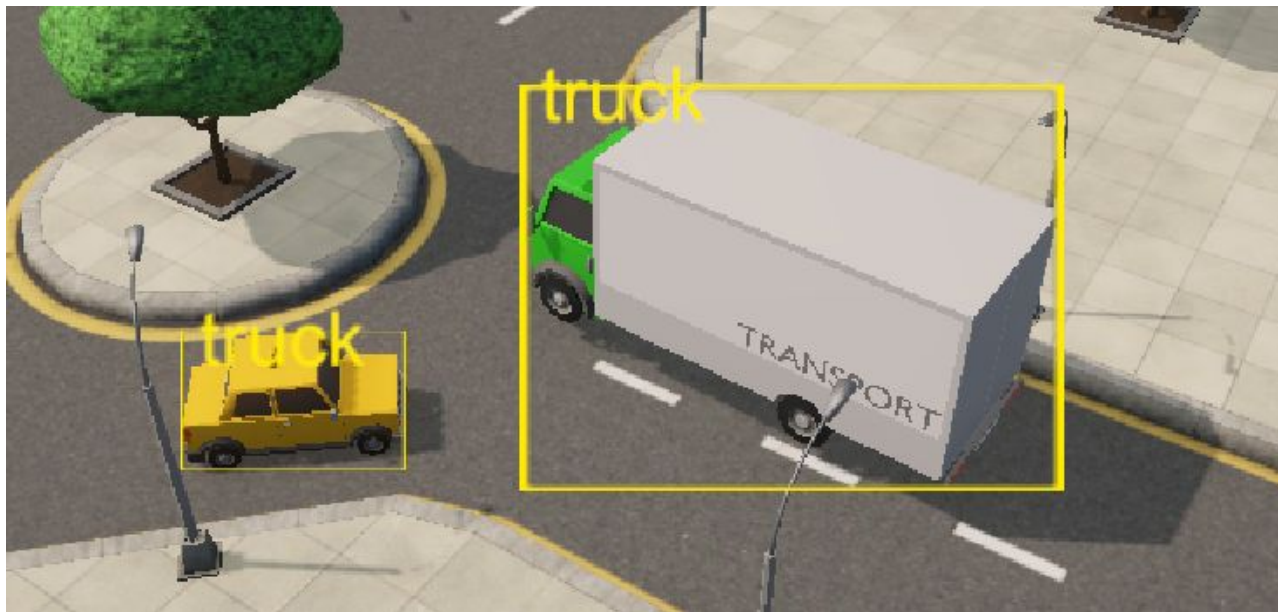
Graph:

The person is to the left of the car
The truck is to the left of the car
The truck is to the left of the person

Demo

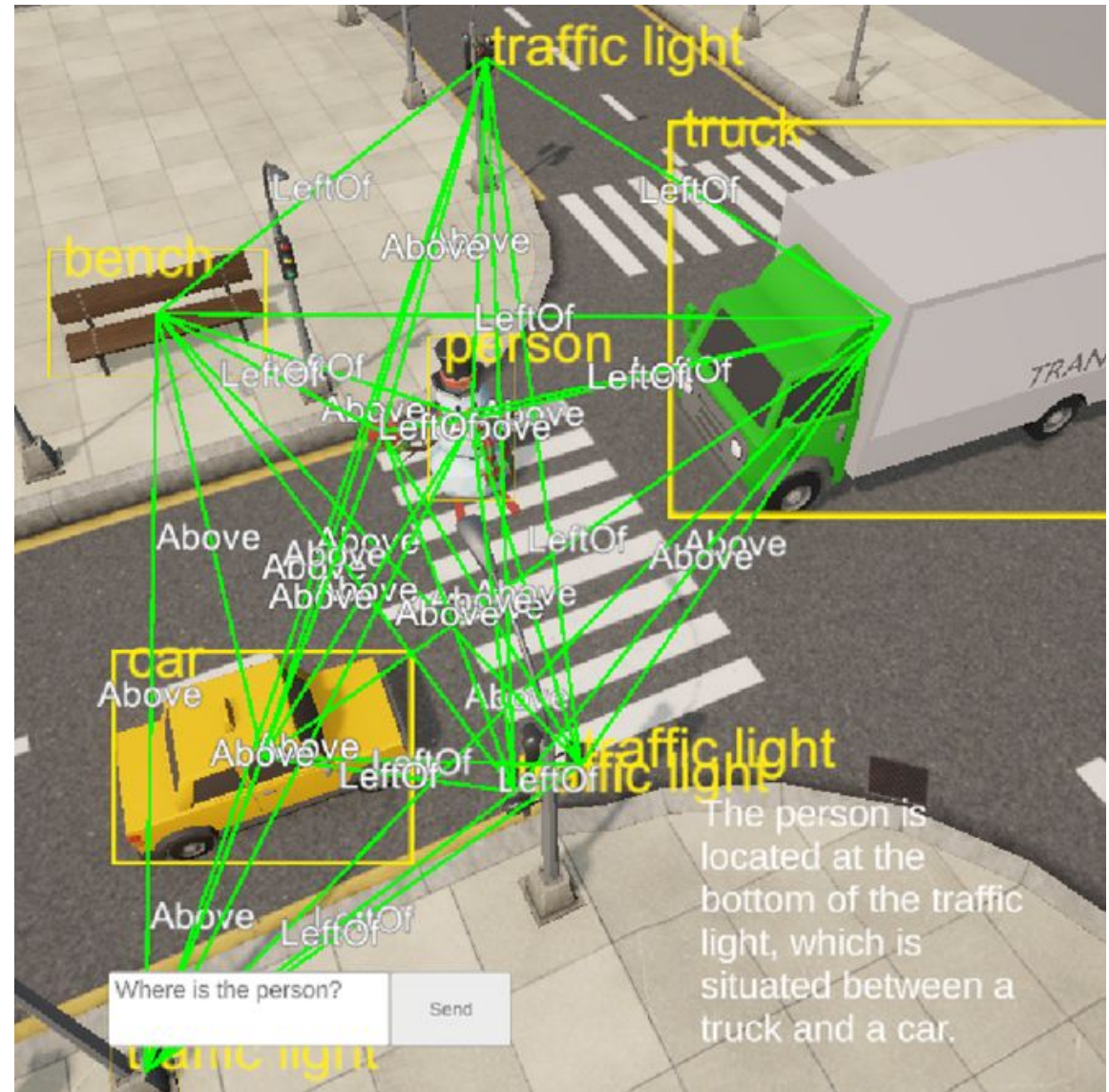
Results

- Detection and labeling issues



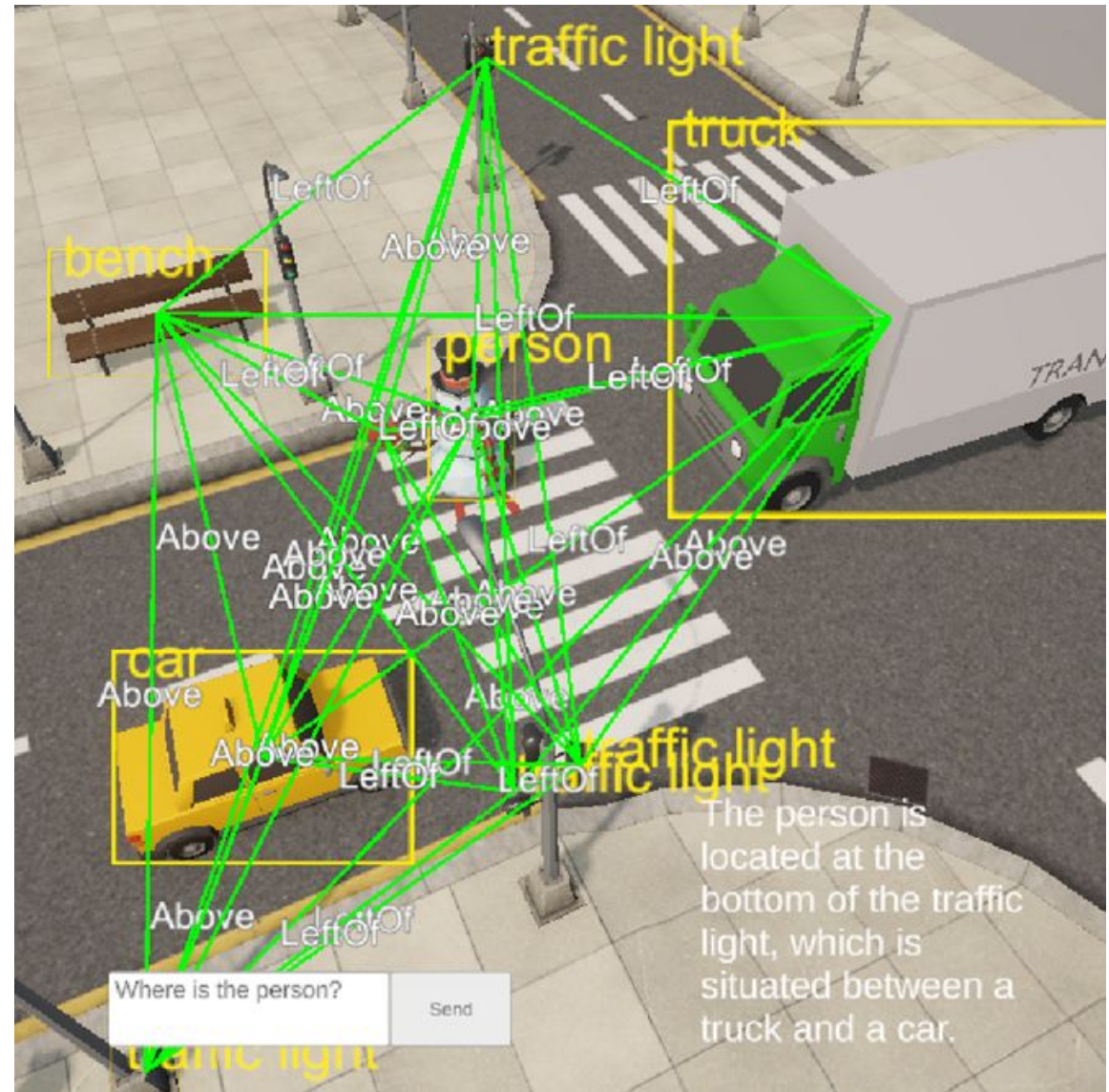
Results

- Detection and labeling issues
- Accurate LLM descriptions, even in complex scenes



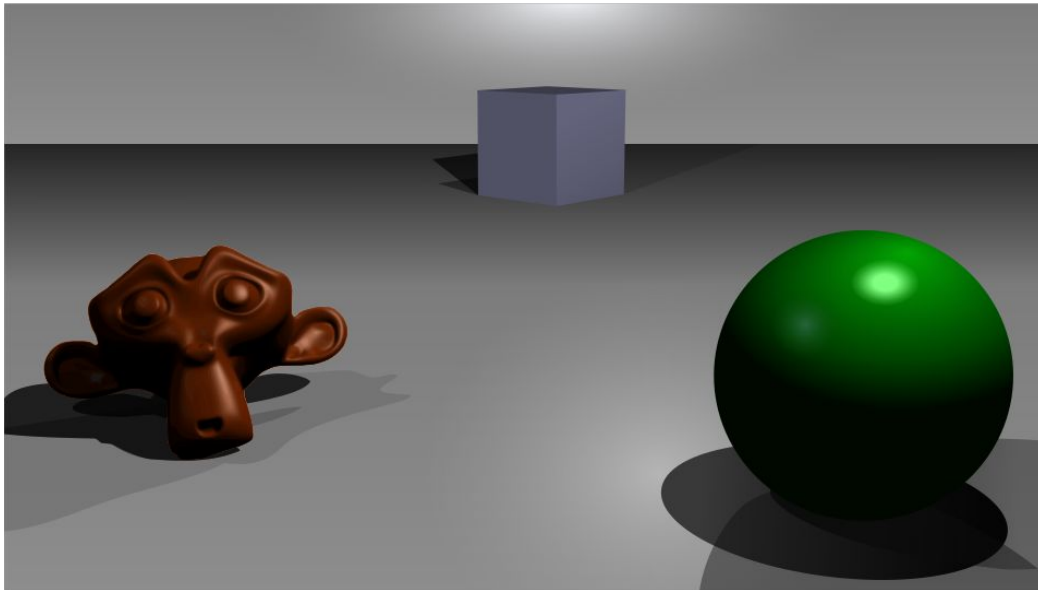
Results

- Detection and labeling issues
- Accurate LLM descriptions, even in complex scenes
- Limited depth comprehension



Future work

- Include depth information (RGB-D)



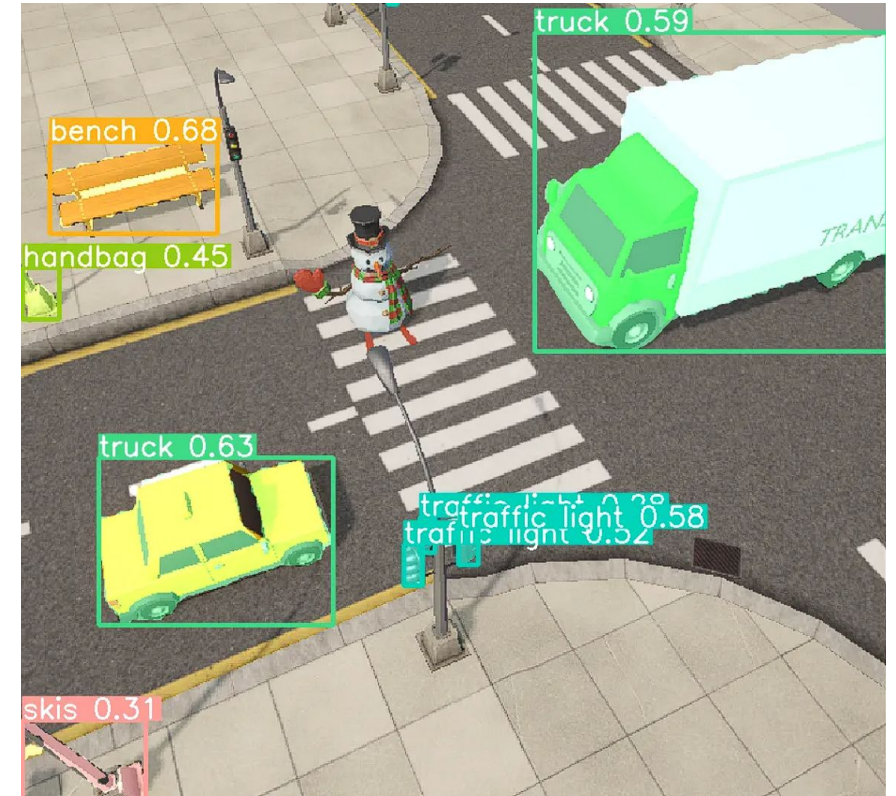
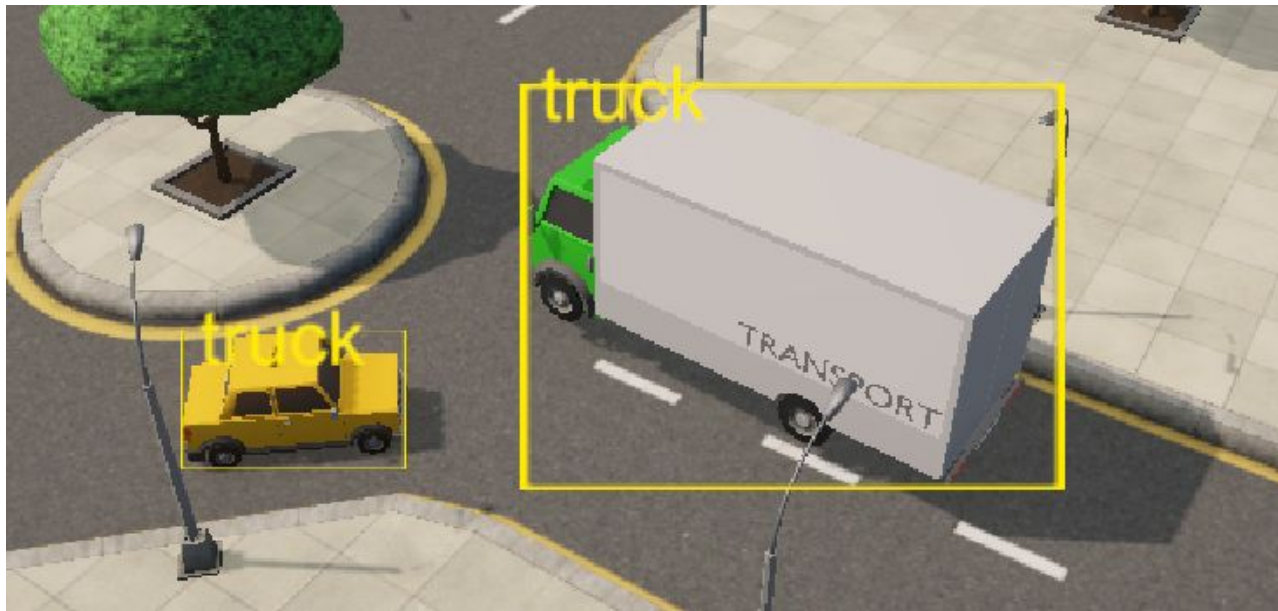
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Future work

- Include depth information (RGB-D)
- Use more relations and more sophisticated heuristics

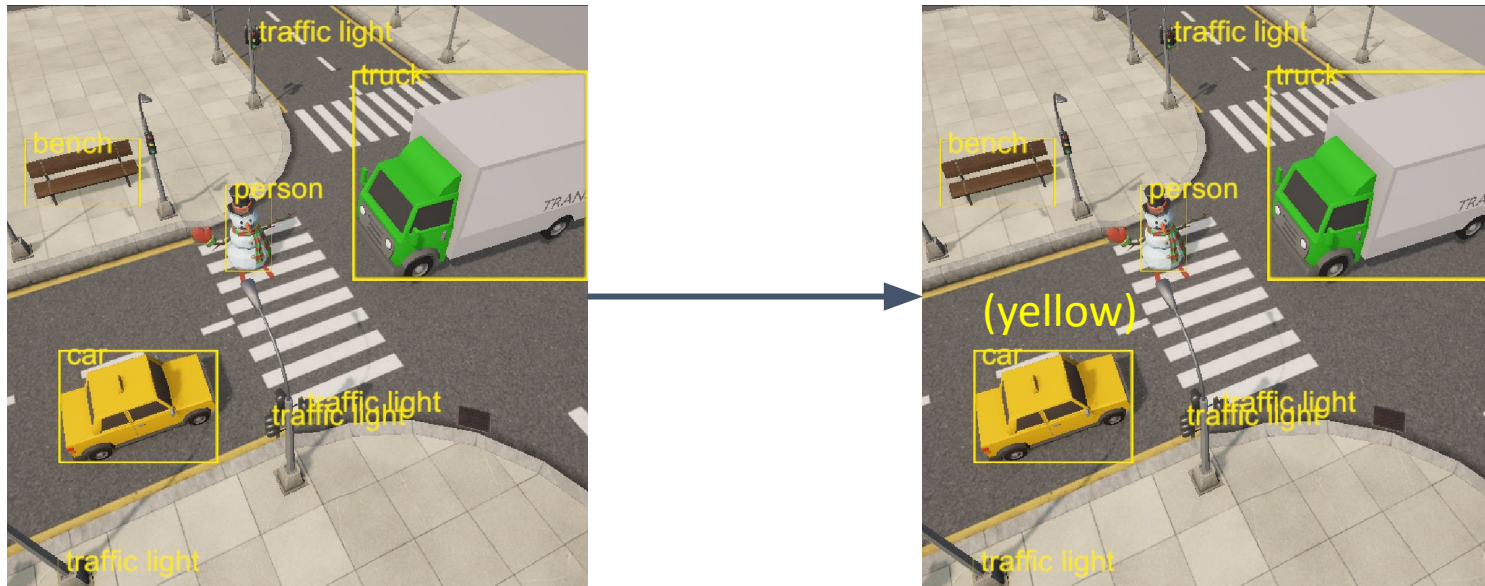
Future work

- Include depth information (RGB-D)
- Use more relations and more sophisticated heuristics
- Use better YOLO models (larger + segmentation)



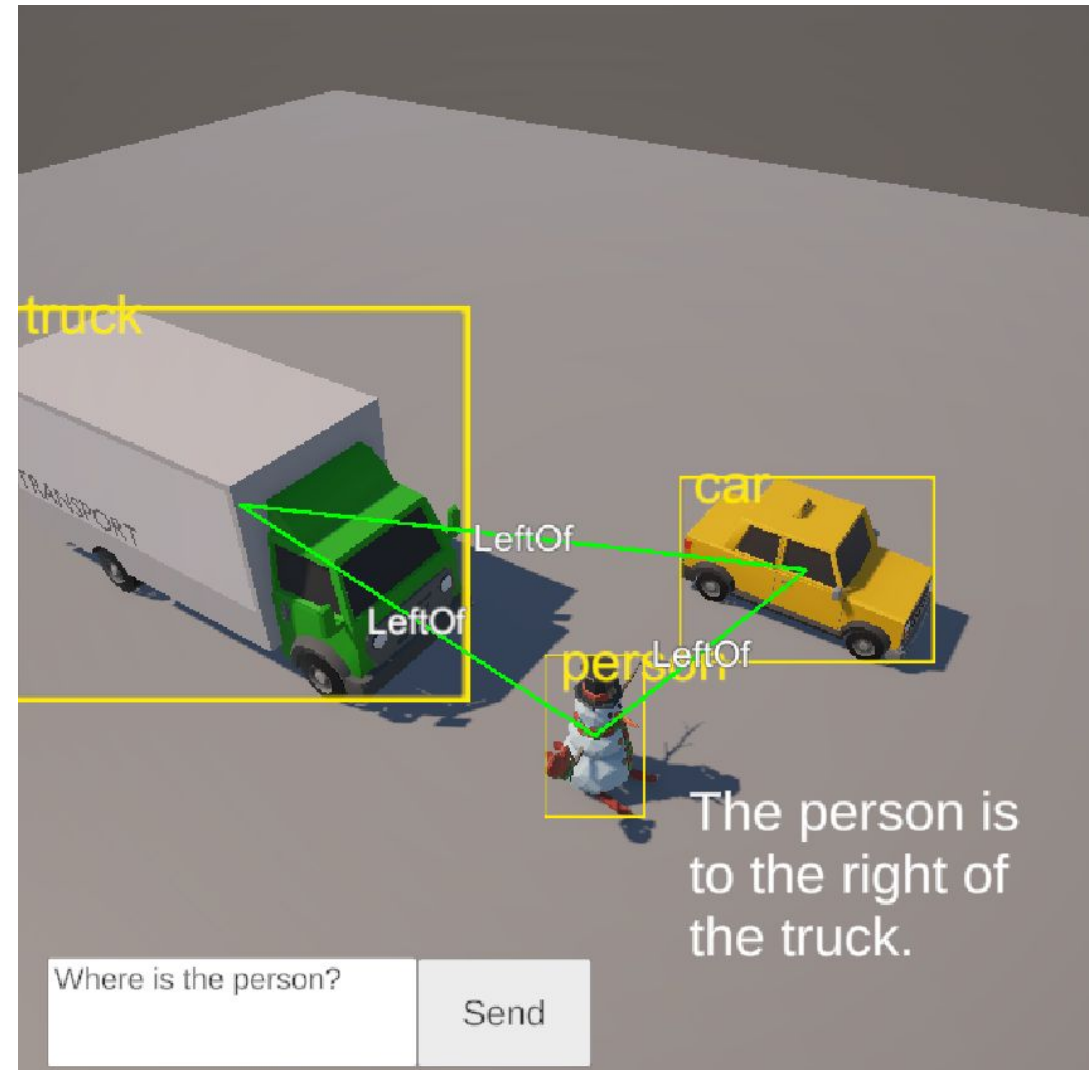
Future work

- Include depth information (RGB-D)
- Use more relations and more sophisticated heuristics
- Use better YOLO models (larger + segmentation)
- Add defining features to detected objects to improve broad labels



Conclusion

- Scene to text descriptions
- Object detection (YOLO)
- Relationship graph (hand-crafted)
- Textual Q&A (LLM)



References

- [1] Julian Müller-Huschke, Marcel Ritter, and Matthias Harders. 2022. Scene synthesis with automated generation of textual descriptions. In Eurographics.
- [2] Zhijie Yan, Shufei Li, Zuoxu Wang, Lixiu Wu, Han Wang, Jun Zhu, Lijiang Chen, and Jihong Liu. 2025. Dynamic open-vocabulary 3d scene graphs for longterm language-guided mobile manipulation. IEEE Robotics and Automation Letters.
- [3] Dave Zhenyu Chen, Ali Gholami, Matthias Nießner, and Angel X. Chang. 2020. can2cap: Context-aware dense captioning in rgb-d scans. 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 3192–3202.
- [4] Xinyan Yang, Fei Hu, Long Ye, Zhiming Chang, and Jiyin Li. 2022. A system of configurable 3d indoor scene synthesis via semantic relation learning. Displays, 74:102168.
- [5] Ting Yao, Yingwei Pan, Yehao Li, and Tao Mei. 2018. Exploring visual relationship for image captioning. In Proceedings of the European Conference on Computer Vision (ECCV).

Thank you for your attention!

- Scene to text descriptions
- Object detection (YOLO)
- Relationship graph (hand-crafted)
- Textual Q&A (LLM)

